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Colony size and turbulent flow speed modulate the calcification response of the coral *Pocillopora verrucosa* to temperature

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Abstract

For poikilothermic organisms in aquatic environment, temperature strongly affects metabolism and modulates the extent to which it is mass transfer limited. The effects of temperature on metabolism are translated into organism performance depending on the organism size, morphology, and the flow regime to which they are exposed. Environmental forcing of metabolism is particularly germane to tropical reef corals, because they increasingly are exposed to upward temperature perturbations to which they display variable responses that remain incompletely understood. In this study, we tested the effects of colony size and seawater turbulence on the response of the common Indo-Pacific branching coral, *Pocillopora verrucosa*, to different seawater temperatures. Using whole-colony calcification as a response variable, 12 tanks (each 150 l) were used in two trials lasting 14 days to contrast the effects of seawater turbulence (two levels) and temperature (25.5 vs 29.5 °C) on colonies varying in size from ~ 4 to 13-cm diameter. Turbulence in the tanks was measured as the root mean square (rms) turbulent flow speed (q_{rms}), which quantified the variation in speed of complex and non-linear flow, regardless of direction. Treatments contrasted low ($q_{rms} = 0.63$ cm s^{-1}) and high ($q_{rms} = 2.07$ cm s^{-1}) turbulence that were ecologically relevant for shallow coral reefs. The effects of turbulent flow speed and temperature on calcification (mg day^{-1}) differed among colony sizes: at 25.5 °C, calcification increased with tissue surface area at a slower rate under high, compared to low turbulent flow speeds; at low turbulent flow speeds, calcification increased with tissue surface area more slowly at 29.5 versus 25.5 °C. These results show that the response of branching corals to temperatures depends on colony size and turbulent flow speed. This outcome suggests that spatial heterogeneity in turbulent flow regimes across a coral reef may provide opportunities for branching corals to exploit colony size to modulate susceptibility to rising temperature.

Introduction

Most shallow coral reefs are strongly influenced by the movement of seawater driven by waves and tides (Monismith 2007). One means by which this influence is exerted is through the thickness of benthic boundary layers (Monismith 2007) that mediates the flux of particles, metabolites, and heat between organisms and seawater (Gaines and Roughgarden 1985; Patterson 1992a, b; Leichter et al. 1998; Jimenez et al. 2008). Despite a great deal of research on the effects of flow on corals and coral reefs (Chamberlain and Graus 1975; Patterson and Sebens 1991; Lowe and Falter 2015; Chan et al. 2016), and advances in elucidating the physical oceanography and biomechanics underlying these effects (Monismith 2007; Chang et al. 2009; Lowe and Falter 2015; Chan et al. 2016), it remains common for coral metabolism to be measured in chambers fitted just with a spinning bar (e.g., Hoogenboom et al. 2010), or in simple light and dark bottles (Baker et al. 2015) (cf. Patterson and Sebens 1991; Finelli et al. 2006). This is unfortunate, as descriptions of the effects of ecologically relevant environmental conditions on scleractinians—including seawater temperature, flow regimes, and light intensity—are critical to project the response of coral reefs to future climate conditions (Pandolfi et al. 2011; Madin et al. 2012).

With strongly negative responses of reef corals to the most recent wave of seawater warming in 2015–2016 (Hughes

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et al. 2017), it is clear that understanding the response of scleractinian corals to temperature needs to be strengthened. Despite a century of studying the effects of temperature on corals (Mayer 1917; Coles and Jokiel 1977), potentially important factors mediating their response remain poorly known, notably in terms of colony size and the turbulent flow regime to which they are exposed. A better understanding of these factors might help reconcile inconsistencies among studies that have shown, for example, that seawater flow was positively associated with bleaching in Mauritius (McClanahan et al. 2005), even though corals best survived the 1998 bleaching in Okinawa when they were in high flow habitats (Nakamura and van Woesik 2001), and were less affected by thermal bleaching when experimentally exposed to high flow speeds (Nakamura and van Woesik 2001; Nakamura et al. 2003; Finelli et al. 2006). Recently, we described the effects of colony size in determining metabolic responses of the branching coral *Pocillopora verrucosa* to temperature and pCO_2 (Edmunds and Burgess 2016), and suggested that these effects could reflect the ways by which colonies of varying size differentially interacted with a constant flow speed of 8 cm s^{-1}.

For corals and coral reefs, rapid unidirectional flow typically accelerates physiological processes in an asymptotic or threshold relationship (Patterson and Sebens 1989, 1991; Thomas and Atkinson 1997; Carpenter and Williams 2007). However, the flow of seawater over coral reefs rarely is unidirectional, and the metabolic response of benthic taxa to ambient flow regimes is mediated by the nuances of flow around organisms and objects differing in sizes and shapes (Monismith 2007; Chang et al. 2009; Lowe and Falter 2015). Indeed, the oscillatory, wave-driven flows that characterize many shallow reefs have stronger positive effects on coral metabolism than unidirectional flow (Falter et al. 2005; Reidenbach et al. 2006). Moreover, for branching corals, the mass transfer implications of flow speed are mediated by the diameter and spacing of branches (Falter et al. 2005; Chang et al. 2009) and, in general, penetration of seawater among them is more effectively accomplished by oscillatory versus laminar flow (Lowe et al. 2005; Reidenbach et al. 2006). In part, this is because moving seawater interacts with roughness elements (i.e., branches) through non-local (created by upstream roughness elements) and local (created by the morphology of specific colonies) turbulence (Falter et al. 2007).

The effects of non-local turbulence on the metabolism of branching corals depend on the extent to which the flow-mediated pressure gradient allows water to move among branches (Falter et al. 2007), versus around and over the colony (Monismith 2007). Low flow speeds favor reduced mass transfer in the cavities between closely spaced branches, but non-local turbulence can increase the exchange of seawater within these spaces for isolated colonies (Falter et al. 2007). Corals with closely spaced branches should be more susceptible to variation in non-local turbulence (Reidenbach et al. 2006), although the reverse is predicted in unidirectional flow (Chang et al. 2014). For branching corals in which branches become longer and wider with increasing colony size, bigger colonies might be more sensitive to changes in non-local turbulence. Further, the ability of seawater flow to penetrate among coral branches is dampened by low frequency variation in orbital wave velocity and closely spaced branches (as described by the Keulegan–Carpenter number Monismith 2007; Chang et al. 2014), as might be found in a large colony with a high density of branches in a habitat affected by wave-driven flow. The metabolic implications of local and non-local turbulence are complicated by the potential for interactive effects between turbulence and the complex ways by which branching corals respond to temperature as a function of colony size (Edmunds and Burgess 2016).

The objective of this study was to extend our previous investigation (Edmunds and Burgess 2016) using the same species of branching coral to address the outstanding question of whether turbulent flow speed can modulate the calcification response (mg day^{-1}) of colonies differing in size (tissue surface area, cm^2) to temperature. The calcification of corals can increase with flow (Dennison and Barnes 1988; Schutter et al. 2010; Goldenheim and Edmunds 2011; Lenihan et al. 2015), and temperatures closer to the threshold value of the thermal performance curve (Pratchett et al. 2015), but how these effects scale with colony size (i.e., Edmunds and Burgess 2016) and respond to ecologically relevant flow regimes remains unknown. Two turbulence regimes were contrasted that are representative of shallow coral reefs (Falter et al. 2007; Monismith 2007), and to achieve this contrast, six tanks (150 l) were used to replicate each turbulence regime. This design differs from most studies of the effects of flow on coral reefs and corals, in which flumes typically are used to create unidirectional and controlled flow regimes (Atkinson and Bilger 1992; Nakamura and van Woesik 2001; Comeau et al. 2014). Our design was rationalized by the objectives of gaining insight into the functional biology of branching corals under replicated ecologically relevant flow conditions, which typically are turbulent among coral colonies on natural reef surfaces (Falter et al. 2007; Monismith 2007).

Materials and methods

Overview

Treatments were applied to twelve tanks (each 150 l), with three tanks allocated to each of the four treatments (treatments described below). The experiment was repeated twice, with each trial lasting 14 days and the second starting the

day the first ended using a different set of coral colonies. Pumps were used to create seawater flow, and flow speeds were representative of those recorded at 10-m depth on the outer reef of Moorea, from where the corals were collected. In situ flow speeds were recorded using an acoustic doppler current profiler (Workhorse Sentinel, Teledyne, RD Instruments, San Diego, CA, USA) (Washburn 2016), and flow in a representative tank was measured with an acoustic doppler velocimeter. Temperature treatments targeted 25.5 °C (L_{temp}) and 29.5 °C (H_{temp}) to span the annual range of seawater temperature on the outer reef of Moorea (Leichter 2015), and when the experiment was conducted (April 2016), ambient seawater temperature was close to the upper value (29.3 ± 0.1 °C [mean ± s.e.m., $N = 30$ daily records from a Hobo U22, ± 0.2 °C accuracy]). One small (~ 5-cm diameter) and one large (~ 10-cm diameter) colony of the common Indo-Pacific coral *Pocillopora verrucosa* was placed in each tank, and at the end of the experiment the size of the corals was measured as tissue surface area. This species was selected for study as it is one of the two most abundant corals on the outer reef of Moorea (Edmunds and Leichter 2016; Tsounis and Edmunds 2016), and was the subject of our previous experiment (Edmunds and Burgess 2016). Whole-colony calcification was measured by buoyant weighing (Spencer-Davies 1989) at the start and end of each experiment as described below. Sample sizes were determined by the results of previous experiments (Edmunds and Burgess 2016).

Coral collection and preparation

Colonies of *Pocillopora verrucosa* on the outer reefs of Moorea were consistently identified by classic morphological features (after Veron 2000). Given the potential for morphologically similar corals to contain multiple genetic groups in this genus (Marti-Puig et al. 2014), we cannot exclude the possibility that our experiment included genetically heterogeneous colonies (see Edmunds et al. 2016). With random allocation of colonies to tanks and treatments, any effects attributed to genetic identity would not bias the results, but they could inflate the residual error, although our previous experimental work with this coral indicated that this effect was not an impediment to detecting effects of experimental treatments (Edmunds and Burgess 2016). Colonies were collected from 10-m depth at two sites, 3-km apart on the north shore, and were pooled between sites for use in the experiment. Colonies in two size classes were selected, one representing colonies 2–3 years old, ~ 5-cm planar diameter, and one representing colonies ~ 10-cm planar diameter, which was close to the largest found on the outer reef in April 2016. The study reef was damaged by an outbreak of the corallivorous seastar *Acanthaster planci* from ~ 2006–2010 (Kayal et al. 2012), and a cyclone in 2010, but by 2016, the coral community on the outer, northshore reef, largely had recovered (Bramanti and Edmunds 2016). As the *P. verrucosa* sampled in 2016 were < 6 years old, they had not achieved the largest size possible for this species (~ 50-cm diameter Veron 2000).

The first group of 12 smaller and 12 larger corals was collected on 8 April 2016, transported to the Richard B Gump South Pacific Research Station, attached to numbered plastic bases using epoxy (Z-Spar A788, Kop-Coat Marine Group, Rockaway, NJ, USA), and placed in an acclimation tank to adjust to laboratory conditions. The acclimation tank was maintained at 28.4 ± 0.2 °C (mean ± s.e.m., $N = 23$) and illuminated on a 12:12 h light:dark photoperiod with LED lamps (SOL White, 6500°K, Aqua-Illumination, Ames, IA, USA) that were adjusted to a target intensity similar to the maximum intensity recorded at the collection depth (~ 645 µmol photons m^{-2} s^{-1} recorded on 10 April 2007). Equipment malfunctions resulted in a mean maximum light intensity in the acclimation tank that was lower than this value (411 ± 9 µmol quanta m^{-2} s^{-1} (± s.e.m., $N = 28$)), although light intensities in the treatment tanks were closer to this target value (~ 530 µmol quanta m^{-2} s^{-1}, described below in results). Light intensities were measured with a spherical quantum sensor (LI-193) attached to a light meter (LI-1400) (Li-Cor Biosciences, Lincoln, NE, USA). The lamps were programmed to increase from 0 to 100% over the first 4 h of each day, to maintain maximum intensity for 4 h, then decline over the final 4 h of the day. Seawater flowed into the acclimation tank at ~ 10 l min^{-1}, and was pumped from ~ 8-m depth in Cook's Bay and filtered through sand. The floor of the acclimation tank (where the corals were located) rotated at two revolutions day^{-1} to remove effects attributed to the position of corals in the tank and asymmetry of lighting and seawater in-flow. The maintenance of constant temperature, light, and flow conditions in the acclimation tank allowed the corals to adjust to common conditions before the experiment began. The seawater temperature was similar to the high temperature treatment used in the subsequent incubations, as well as the ambient seawater temperature when the experiment was conducted.

The first group of corals was acclimated for 5 d before being buoyant weighed and randomly allocated to the 12 tanks on 13 April 2016, with one small and one big coral in each tank. The smaller coral was placed on a 4-cm high PVC pedestal that supported it above the tank floor at a height similar to that of the larger coral. Corals were maintained in these tanks for 14 days, and were moved daily to prevent potentially confounding effects attributed to their position in the tanks. At the conclusion of the first experiment (27 April 2016), the corals were buoyant weighed, and the change in buoyant weight converted to dry mass of $CaCO_3$ using the density of aragonite (2.93 g cm^{-3}). Finally, the corals were dried and the tissue surface area was determined by wax

dipping (Stimson and Kinzie, 1991), and the values were used to standardize calcification to time (mg day^{-1}).

The second trial began the day the first trial ended, and was conducted in an identical manner. Corals for the second trial were collected on 21 April 2016, acclimated until 27 April, buoyant weighed, and placed into the treatment tanks from which the first batch of corals had been removed. The second trial ran until 11 May 2016, when the corals were weighed and measured for surface area as described above.

Treatments

The treatments involved factorial combinations of two temperatures and two flow treatments. The 12 tanks were independently regulated for temperature (using a chilling loop and heater) and light, and received a constant flow of seawater from ~ 8-m depth in Cook's Bay and filtered through sand. The lights were identical to those used in the acclimation tank, and supplied light with the same photoperiod. The combinations of temperatures and turbulence were randomly allocated among tanks (with three tanks for each of the four combinations). Turbulent flow speeds were manipulated using one (low turbulence, L_{turb}) or two (high turbulence, H_{turb}) pumps (Aqua-Mag 72,650 l h^{-1}, Danner Manufacturing, Inc., Islandia, NY, USA) in each tank (Fig. 1), with the second pump attached to a capped, PVC pipe (13-mm inside diameter) with holes (8-mm diameter) that discharged seawater (in a design similar to Falter et al. 2007). An acoustic doppler velocimeter (ADV, Vectrino) (Nortek, Boston, MA, USA) was used to quantify flow, but since this tall instrument would not fit in the tanks while they were tightly arrayed in the mesocosm racks, flow was measured in an identical tank set up on a bench and containing paired corals (one small and one large) and the same pump configurations as used in the mesocosm. We assume that slight differences among the tanks in location of the pump, heater, and standpipe had negligible effects on the complex turbulent flow created by the pumps. The ADV operated at 25 Hz and recorded flow in x, y, and z directions that were used to calculate the turbulent flow speed (q, Eq. 1), and its root-mean-square (rms) average value (q_{rms}) (after Falter et al. 2007):

$$q = \sqrt{1/3(x^2 + y^2 + z^2)}. \quad (1)$$

Measurements of water motion were taken from seven positions around the large colonies, and three positions around the small colonies. Turbulent flow speed (q) and its rms average (q_{rms}) were compared between treatments, and between corals in the same tank. As the 12 tanks were identical, it was reasonable to assume that turbulent flow speed in the L_{turb} and H_{turb} tanks was similar to that in the one tank configured on the bench to match the L_{turb} and the H_{turb} design.

Conditions in the tanks were monitored throughout the trials. Seawater temperature was measured with a certified digital thermometer (± 0.05 °C certified, model 15-077, Fisher Scientific, Pittsburgh, PA, USA), initially at multiple times every day, but daily when conditions had stabilized. Light intensity was measured close to noon every day (when LEDs were at maximum output) using a Li-Cor 4-π quantum sensor (LI-193) positioned at the depth of the corals in each tank. The rate of fresh seawater flowing into the tanks was recorded daily using a measuring cylinder (ml s^{-1}). Seawater chemistry (pH_T, P_{CO_2}, A_T, salinity, and Ω_{arag}) was measured once in each trial (using standard procedures in Dickson et al. 2007) to qualitatively evaluate changes in seawater conditions resulting from coral metabolism. The analysis of seawater chemistry was not designed to support a statistical contrast of conditions among tanks or temperature

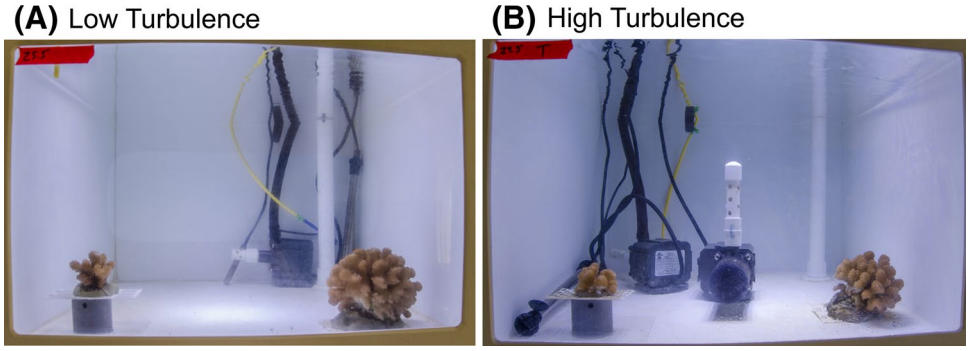

Fig. 1 Photographs of representative tanks in low turbulence (L_{turb}) **a** and high turbulence (H_{turb}) **b** treatments showing a small and large colony of *Pocillopora verrucosa* in each tank. Tanks were 55-cm wide, and in the L_{turb} treatment, turbulence was created by one pump (back of tank) that circulated water through a cooling coil. In the H_{turb} treatment, a second pump (center) circulated seawater through a capped pipe, drilled with holes (~ 8-mm diam.). Tanks received a continuous supply of seawater (~ 180 ml min^{-1}) that overflowed into a standpipe (back, right), and were bubbled with compressed air. Corals were moved within the tanks daily to prevent position effects, and rms turbulent flow speed (q_{rms}) was characterized in an identical tank into which the ADV could be inserted

treatments, rather to confirm that the turnover of seawater in the tanks prevented build up of CO_2 due to coral metabolism at night, and depletion of A_T during the day due to coral calcification during the day. Replicate analyses of seawater chemistry were not completed because this experiment did not include P_{CO_2} as a treatment condition, and because previous work (in which P_{CO_2} was manipulated) showed that control P_{CO_2} conditions (as in all tanks in the resent study) created seawater chemistry that was stable among tanks and over time (Edmunds and Burgess 2016).

Analyses

Treatment conditions of seawater temperature and light intensity were compared among treatment combinations in the two trials using nested, Model III ANOVA in which treatment was a fixed effect, and tank was a random effect nested in treatment. q and q_{rms} were compared between turbulent flow treatments and small and large corals using t tests.

The objective of this study was to estimate the effects of turbulent flow speed and temperature on the relationship between colony size (tissue area) and whole-colony calcification (mg day^{-1}). Therefore, the statistical approach involved fitting parametric models and describing the extent to which the quantitative relationships (represented by parameters for the slopes and intercepts) between calcification rate and size differed among turbulent flow speeds and temperatures (after Edmunds and Burgess 2016). Bayesian linear mixed effects models were used, which provide the most accurate estimates of mixed model parameters and p values for small datasets. Tissue surface area was modeled as a continuous predictor, and turbulent flow speed and temperature were modeled as categorical predictors (each with two levels) (an ANCOVA-like design). Trials ($N = 2$) and tanks-within-trials ($N = 3$, for each treatment combination in each trial) were treated as random effects. Tissue surface area was mean standardized for analysis (i.e., the overall mean coral size was subtracted from each individual coral size), so that the intercept parameter represents the expected calcification rate of an averaged-size coral (which was 295 cm^2 in our sample), rather than a coral of size 0 cm^2 (Table 2). This method of standardization does not change the interpretation of the slope parameters. Deviance Information Criterion (DIC) was used to compare the fit of models with and without the three-way interaction between size, temperature, and turbulence.

Models were fit to data using Markov Chain Monte Carlo (MCMC) methods in the R package MCMCglmm (Hadfield 2010). Flat (uninformative) inverse-Wishart priors were used. Since the group-level variance (i.e., the estimate of variance due to trials and tanks) was close to zero, a parameter expansion method was used to improve the mixing properties of the MCMC chain and provide a better description of the posterior distribution (following Hadfield 2014). In all models, MCMC chains were run for 10^6 iterations with a burn-in of 3000 and a thinning interval of 10. The mixing properties of all parameters were checked to ensure no systematic trends, and the sensitivity of results to alternative prior specifications was also checked. Regression coefficients are reported with 95% Bayesian credible intervals, which provide the interval within which there is a 95% probability that the mean will occur.

Results

Treatments

Treatment conditions were effectively maintained in the tanks (Table 1). In Trial 1, treatments contrasted a mean temperature of 25.4 ± 0.1 °C with 29.5 ± 0.1 °C (± s.e.m., $N = 6$), at a mean light intensity of 545 ± 8 µmol quanta m^{-2} s^{-1} (± s.e.m., $N = 12$), with seawater flowing into the tanks at 171–184 ml min^{-1}. In Trial 2, the treatments contrasted a mean temperature of 25.3 ± 0.1 °C with 29.4 ± 0.1 °C (± s.e.m., $N = 6$), at a mean maximum light intensity of 526 ± 7 µmol quanta m^{-2} s^{-1} (± s.e.m., $N = 12$), with seawater flowing into the tanks at 195–214 ml min^{-1}. Seawater chemistry (pH_T, A_T, P_{CO_2}, Ω_{arag}, and S) qualitatively was similar in all tanks and between trials. Although only one measurement was taken for each tank, the 12 measurements suggested there was no depletion of A_T caused by calcification of the corals, or elevation of P_{CO_2} relative to open seawater caused by nocturnal respiration.

Turbulent flow speeds differed between turbulence treatments (Fig. 2), and in L_{turb}, maximum flow speed was ~ 0.4 cm s^{-1}, minimum flow speed was ~ 0.3 cm s^{-1}, and the mean flow speed was ~ 2.8 cm s^{-1}. In H_{turb}, the maximum flow speed was ~ 26.5 cm s^{-1}, minimum flow speed was ~ 1.4 cm s^{-1}, and the mean flow speed was ~ 6.5 cm s^{-1}. In L_{turb}, mean flow speeds ranged from 0.5–1.1 cm s^{-1} around small corals, and 1.0–3.2 cm s^{-1} around large corals, which correspond to turbulent flow speeds of 0.22–0.67 and 0.44–1.01 cm s^{-1}, respectively; at H_{turb}, mean flow speeds ranged from 3.2 to 4.2 cm s^{-1} around small corals, and 3.6–5.3 cm s^{-1} around large corals, which correspond to turbulent flow speeds of 1.71–2.44 cm s^{-1} and 1.51–3.07 cm s^{-1}, respectively (Fig. 2). The mean flow speed ($\bar{U}$) was 189% greater in H_{turb} versus L_{turb} ($t = 7.586$, $df = 18$, $P < 0.001$), and was higher around big versus small corals in L_{turb} ($t = 2.363$, $df = 8$, $P = 0.046$) but not H_{turb} ($t = 1.811$, $df = 8$, $P = 0.108$); q_{rms} was 204% greater in H_{turb} versus L_{turb} ($t = 8.178$, $df = 18$, $P < 0.001$), but was similar between small and larger corals ($t \leq 1.965$, $df = 8$, $P \geq 0.085$) (Fig. 2). Mean flow speed in the tanks was ecologically relevant

Table 1 Summary of conditions in 12 tanks during Trial 1 and 2 in which colonies of *Pocillopora verrucosa* were exposed to combinations of temperatures and rms turbulent flow speeds (q_{rms})

Trial	Tank#	Treatment	pH_T	A_T (μmol kg^{-1})	P_{CO_2} (μatm)	Ω_{arag}	T (°C)	S	Light (μmol m^{-2} s^{-1})	Flow rate (ml min^{-1})
1	1	H_{temp}–H_{turb}	8.03	2309	405	3.9	29.46 ± 0.04	35.6	556 ± 10	173 ± 14
	2	L_{temp}–L_{turb}	8.06	2302	375	3.6	25.30 ± 0.03	35.6	572 ± 6	176 ± 14
	3	H_{temp}–H_{turb}	8.02	2312	417	3.9	29.45 ± 0.05	35.6	531 ± 6	175 ± 16
	4	L_{temp}–L_{turb}	8.07	2303	365	3.7	25.37 ± 0.04	35.6	587 ± 15	171 ± 14
	5	L_{temp}–H_{turb}	8.06	2317	380	3.6	25.61 ± 0.32	35.6	510 ± 8	176 ± 14
	6	H_{temp}–L_{turb}	8.03	2317	412	3.9	29.84 ± 0.04	35.6	529 ± 4	172 ± 13
	7	L_{temp}–H_{turb}	8.06	2305	377	3.6	25.17 ± 0.03	35.6	579 ± 6	171 ± 14
	8	H_{temp}–H_{turb}	8.04	2302	401	3.9	29.47 ± 0.04	35.6	556 ± 8	172 ± 13
	9	H_{temp}–L_{turb}	8.03	2328	417	3.9	29.66 ± 0.15	35.6	571 ± 12	175 ± 12
	10	H_{temp}–L_{turb}	8.03	2317	413	3.8	29.30 ± 0.05	35.6	518 ± 7	172 ± 13
	11	L_{temp}–H_{turb}	8.06	2302	380	3.6	25.38 ± 0.10	35.6	505 ± 5	184 ± 13
	12	L_{temp}–L_{turb}	8.07	2305	365	3.7	25.76 ± 0.05	35.6	531 ± 4	178 ± 11
2	1	H_{temp}–H_{turb}	8.07	2310	367	4.1	29.36 ± 0.18	35.6	572 ± 18	203 ± 15
	2	L_{temp}–L_{turb}	8.09	2321	350	3.8	25.27 ± 0.08	35.6	533 ± 7	195 ± 9
	3	H_{temp}–H_{turb}	8.01	2324	431	3.8	29.51 ± 0.07	35.6	520 ± 2	203 ± 14
	4	L_{temp}–L_{turb}	8.07	2312	366	3.7	25.30 ± 0.05	35.6	545 ± 15	210 ± 10
	5	L_{temp}–H_{turb}	8.04	2322	401	3.5	25.15 ± 0.05	35.6	494 ± 3	214 ± 10
	6	H_{temp}–L_{turb}	8.04	2328	405	3.9	29.46 ± 0.12	35.6	514 ± 6	208 ± 9
	7	L_{temp}–H_{turb}	8.07	2309	375	3.6	25.14 ± 0.04	35.6	512 ± 11	205 ± 6
	8	H_{temp}–H_{turb}	8.07	2325	366	4.2	29.38 ± 0.05	35.6	559 ± 7	198 ± 13
	9	H_{temp}–L_{turb}	8.07	2311	363	4.1	29.42 ± 0.14	35.6	554 ± 16	204 ± 13
	10	H_{temp}–L_{turb}	8.02	2311	416	3.8	29.25 ± 0.08	35.6	514 ± 2	200 ± 14
	11	L_{temp}–H_{turb}	8.07	2324	368	3.7	25.29 ± 0.10	35.6	495 ± 4	203 ± 11
	12	L_{temp}–L_{turb}	8.08	2319	362	3.7	25.62 ± 0.07	35.6	502 ± 5	209 ± 8

H_{temp}–H_{turb}, high temperature, high turbulence; H_{temp}–L_{turb}, high temperature, low turbulence; L_{temp}–H_{turb}, low temperature, high turbulence, and L_{temp}–L_{turb}, low temperature, low turbulence; pH_T, pH on the total scale; A_T total alkalinity; P_{CO_2}, partial pressure of carbon dioxide; Ω_{arag} aragonite saturation state, S salinity, *Light*, light intensity, and Flow rate = rate at which fresh seawater flowed into each tank. Sample sizes: temperature, $N = 15, 14$; light, $N = 14, 4$; flow, $N = 11, 6$ (all trial 1 and 2, respectively); all others $N = 1$

for 10-m depth on the outer reef of Moorea in April 2015 (where mean flow speed = 8.9 cm s^{-1}, 95% confidence interval = 1.0–21.4 cm s^{-1}). Turbulent flow speeds in the tanks were lower than ~ 6.0–9.5 cm s^{-1} recorded 30-cm above the bottom on the shallow reef flat of Kaneohe Bay, Hawaii (50–100 m shoreward of the surf zone) (Falter et al. 2007), but was higher than ~ 0.5–1.5 cm s^{-1} between colonies of *Porites compressa* in an experimental community exposed to rms flow speeds of 0–15 cm s^{-1} (Falter et al. 2007).

Response of the corals to the treatments

The corals ranged in diameter from 37 to 119 mm in Trial 1, and from 47 to 128 mm in Trial 2, which corresponded to tissue areas of 32–668 cm^2 and 59–738 cm^2, respectively. The corals were a normal light brown color when collected (Fig. 1), and they remained this color throughout the experiments and expanded their polyps at night.

All but one coral increased in mass throughout the experiments, and the one exception (a large coral in the H_{temp}–L_{turb} treatment) was caused by a piece of the skeleton that broke away during handling; this coral was dropped from the analysis. In Trial 1, corals grew between 0.18 and 5.84 g, which corresponded to area-normalized rates of 0.18–0.62 mg cm^{-2} day^{-1}, and an overall increase in skeletal dry mass of 0.4–1.1%. In Trial 2, corals grew between 0.08 g and 3.84 g, which corresponded to area-normalized rates of 0.13–0.72 mg cm^{-2} day^{-1}, and an overall increase in skeletal dry mass of 0.1–0.7%.

The relationship between tissue surface area (cm^2) and whole-colony calcification (mg day^{-1}) differed between the turbulence and temperature treatments (Table 2, Fig. 3). The model fitting different slopes in each treatment combination (Table 2) had a lower DIC than an additive model fitting the same slope of calcification against area in all treatments (ΔDIC = 3.05). Therefore, the effects of turbulent flow speed and temperature on calcification rate (as mg day^{-1})

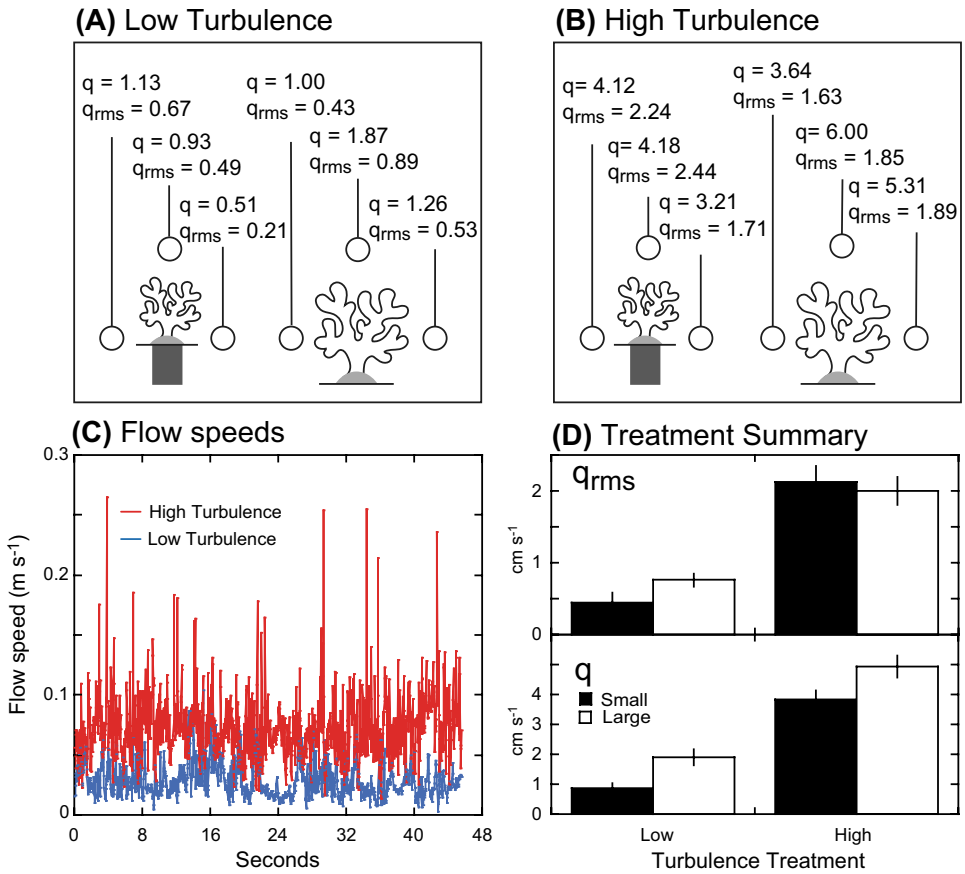

Fig. 2 Summary of turbulent flow speeds created in **a** L_{turb}, and **b** H_{turb}, tanks (Fig. 1). Seawater flow was recorded with an ADV sampling in the *x*, *y*, and *x* planes at 25 Hz. Turbulent flow speed (q) was used to calculate the rms mean turbulent flow speed (q_{rms}, cm s^{-1}), around two coral colonies (7 positions around the big colony, and 3 for the small colony, 3 of which are shown with circles around each colony). **c** Representative flow speeds (q) against time at 15-cm above the floor of the tank in the H_{turb} (red) and L_{turb} (blue) treatments. **d** Summary of treatment conditions for a small and large coral placed in L_{turb} and H_{turb} treatments as evaluated by q and q_{rms} (mean ± s.e.m. shown, with $N = 3$ for small and $N = 7$ for large corals)

Table 2 Parameters (with 95% credible intervals in parentheses) from Markov Chain Monte Carlo (MCMC) Bayesian linear mixed-effects models fit to whole-colony calcification rate (mg day^{-1}) in *Pocillopora verrucosa*

Parameter	Interpretation	Estimate (95% credible intervals)	MCMC *P* value
α	Calcification of corals in L_{temp}–L_{turb} at the mean size of 295 cm^2	**130.18 (101.37–157.96)**	**< 0.001**
β_1	Slope of coral surface area (cm^2) vs. calcification in L_{temp}–L_{turb}	**0.41 (0.30–0.52)**	**< 0.001**
β_2	Difference in calcification of corals in H_{temp}–L_{turb} compared to L_{temp}–L_{turb} at size 295 cm^2	− 41.23 (− 83.24 to 0.45)	0.053
β_3	Difference in calcification of corals in L_{temp}–H_{turb} compared to L_{temp}–L_{turb} at size 295 cm^2	− 31.91 (− 71.27 to 8.34)	0.112
β_4	Change in the slope of coral surface area (cm^2) vs. calcification in H_{temp}–L_{turb} vs L_{temp}–L_{turb}	**− 0.17 (− 0.35 to 0)**	**0.047**
β_5	Change in the slope of coral surface area (cm^2) vs. calcification in L_{temp}–H_{turb} vs L_{temp}–L_{turb}	**− 0.22 (0.39 to 0.06)**	**0.008**
β_6	Difference in calcification of corals in H_{temp}–H_{turb} compared to L_{temp}–L_{turb} at size 295 cm^2	34.91 (− 22.10 to 93.17)	0.225
β_7	Partial regression coefficient in H_{temp}–H_{turb}	**0.31 (0.07 to − 0.54)**	**0.013**
σ^2_α	Variance in calcification among tanks and years	175 (0–714)	
σ^2_y	Residual variance in calcification among corals within each tank	2071 (1101–3170)	

Corals from a range of sizes (tissue surface area, cm^2) were measured at either 25.5 °C (L_{temp}) or 29.5 °C (H_{temp}) and low (L_{turb}) or high (H_{turb}) turbulent flow speed. The model was: $y_i \sim \alpha_{j[i](L_{temp}-L_{turb})} + \beta_1 area_{i(L_{temp}-L_{turb})} + \beta_{2(H_{temp}-L_{turb})} + \beta_{3(L_{temp}-H_{turb})} + \beta_{4area_i(H_{temp}-L_{turb})} + \beta_{5area_i(L_{temp}-H_{turb})} + \beta_{6(H_{temp}-H_{turb})} + \beta_7 area_{i(H_{temp}-H_{turb})} + \epsilon_i$, where $\alpha_j \sim N(\mu_a, \sigma^2_\alpha)$ and $\varepsilon_i = N(0, \sigma^2_y)$. Values in bold indicate parameter estimates that are significantly diferent from zero. Tissue surface area was mean standardized for analysis (mean size = 295 cm^2)

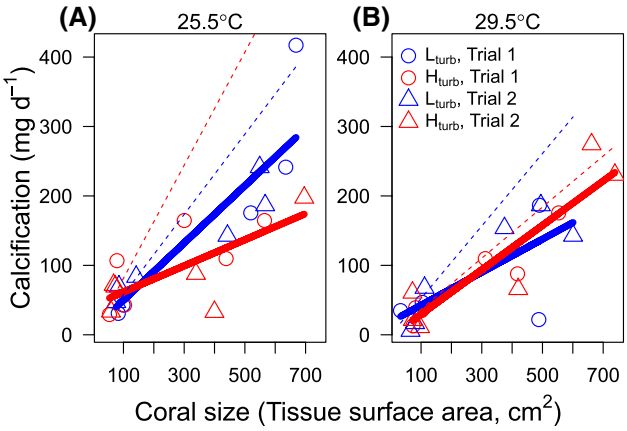

Fig. 3 Whole-colony calcification rate (mg day^{-1}) of *Pocillopora verrucosa* as a function of coral size (tissue surface area, cm^2) at two temperatures (**a**, **b**) and two turbulent flow speeds (blue lines = low turbulent flow speed, q_{rms} = 0.63 cm s^{-1}; red = high turbulent flow speed, q_{rms} = 2.07 cm s^{-1}), based on two 14 d trials (symbols) conducted in April 2016. Each symbol represents one coral colony (Fig. 1). Thick solid lines show the model fits from a Bayesian Linear Mixed Effect model. Parameter estimates are shown in Table 2. The dashed lines show the expected whole-colony calcification calculated by scaling up the area-normalized calcification (i.e., mg cm^2 day^{-1}) for the tissue surface area for the smallest coral sampled (33 cm^2). Deviations from the dashed line indicate the inaccuracy associated with using small corals (e.g., branches and fragments) to estimate whole-colony calcification rates of larger corals, as a result of rates per unit area declining with larger size

were not consistent across the range of coral sizes examined (i.e., colonies ~ 4–13 cm diameter). For example, for every square centimeter increase in tissue area, colony mass increased 0.41 mg day^{-1} at 25.5 °C and low turbulent flow speed (i.e., q_{rms} = 0.63 cm s^{-1}), but calcification increased with tissue area approximately half as fast under the high turbulent flow speed (i.e., q_{rms} = 2.07 cm s^{-1}) (β_5 in Table 2, Fig. 3a). Similarly, at low turbulent flow speeds, calcification increased with tissue area at a slower rate at 29.5 °C compared to that at 25.5 °C (β_4 in Table 2). These relationships indicate that estimates of area-normalized calcification (i.e., units of mg cm^2 day^{-1}) from small coral fragments will overestimate whole coral calcification rates for larger corals (dashed lines versus solid lines in Fig. 3).

Discussion

Despite decades of research on coral physiology, there remains much to learn about their responses to the suite of environmental conditions that are ecologically relevant on coral reefs, including (but not limited to) temperature, flow regimes, light intensity, and pH (e.g., Jimenez et al. 2008; Chang et al. 2009; Rivest and Gouhier 2015; Ainsworth et al. 2016). Here, rather than simply addressing the response of small coral fragments to a single factor (i.e., temperature), we focused on the interactive effects with turbulent flow speeds and colony size. Seawater temperature was manipulated to two levels within the annual range of temperatures in the habitat from which the corals were collected, but avoided extreme high temperatures that cause coral bleaching (Brown and Dunne 2015). Our results show that ecologically relevant temperatures and turbulent flow speeds have different effects on calcification depending on the size of the coral. They confirm, therefore, the fallacy of functional equivalence between small and large corals (Edmunds and Burgess 2016), and reveal a novel way through which turbulent flow regimes modulate the selective consequences of variation in size for branching corals. Spatial heterogeneity in turbulent flow regimes across a coral reef may provide opportunities for branching corals to exploit colony size to modulate susceptibility to rising temperature.

The effects of temperature on coral calcification predicted from current literature did not occur in our experiment. Coral calcification (usually normalized to tissue area, or expressed as linear extension) typically increases with temperature according to an asymmetric response characterized by a threshold temperature and a steep decline in rate at temperatures higher than this value (Edmunds 2005; Castillo et al. 2014; Pratchett et al. 2015). With a positive threshold of ~ 28 °C, and thermal dependency on either side of this value (Pratchett et al. 2015), it was reasonable to have expected that the present experimental design, which contrasted 29.5 °C with 25.5 °C, would have revealed reduced calcification at the lower temperature. This was not the case in our analyses with whole coral colonies (Fig. 3). Likewise, in our previous study that also was conducted with *Pocillopora verrucosa* from the outer reef of Moorea, a 3.2 °C downward temperature excursion from 29.7 °C had no effect on calcification in the same mesocosm system, and under flow conditions corresponding to low turbulence in the present study (Edmunds and Burgess 2016).

The effects of turbulent flow speed on calcification were also not consistent with the effects predicted from analyses that typically have been conducted with well-controlled and unidirectional flow in flumes. Given that corals and coral reefs are mass transfer limited for a variety of metabolic processes at low flow speeds (Lesser et al. 1994; Carpenter and Williams 2007; Mass et al. 2010), and the calcification rate of corals can be increased by high flow speeds (Dennison and Barnes 1988; Schutter et al. 2010; Goldenheim and Edmunds 2011; Lenihan et al. 2015), it was reasonable to expect that calcification in *P. verrucosa* would be increased by enhanced turbulent flow speed. Again, this was not the case.

Parsimonious reasons why our results differ from predictions (described above) are that they reflect the use of colony size as a covariate in the experimental design, and a

focus on a species that produces a densely branched coral-lum (Veron 2000; Edmunds and Burgess 2016). One of the classic assumptions of a colonial modular design, characteristic of most reef-building scleractinians (Jackson 1979; Rosen 1986), is that modularity provides freedom from the constraints of size through isometric scaling (Hughes 2005), where size-specific physiological responses should remain constant across all sizes (but see Burgess et al. 2017). Our results show that temperature and turbulent flow speed alter the extent to which calcification rate of small corals can predict the responses of larger corals to the same conditions (Schmidt-Nielsen 1984; Hughes 2005; Edmunds and Burgess 2016), largely because scaling was not universally allometric or isometric. Although explicit tests of scaling in corals are rare, most empirical studies consider scaling to be either isometric or allometric in any one species (but see Edmunds 2005), rather than being environment dependent (Jokiel and Morrissey 1986; Patterson 1992b; Vollmer and Edmunds 2000; Edmunds and Burgess 2016). Transitioning between isometric and allometric scaling is likely to be particularly common in corals with densely branched coralla (like *P. verrucosa*), where the shaded and flow-restricted interstices of large colonies indicate why they are not simply the sum of multiple small colonies (e.g., Chang et al. 2009, 2014). As a result, whole-colony responses to our experimental treatments were not consistent for a range of coral sizes, and this prevented detecting main effects of temperature and turbulent flow speed.

The present results, combined with our previous study (Edmunds and Burgess 2016), reveal that the effects of temperature on calcification in *Pocillopora verrucosa* depend on colony size. Our previous experiment was conducted at a low turbulent flow speed (created in the same tanks used in the present study, which were arranged in the "L_{turb}" configuration, although turbulent flow speed was not measured), and it employed colonies of broadly the same range of sizes (7–646 cm^2) as used in the present study (Edmunds and Burgess 2016). While the present results replicated the previous outcome for temperature (cf. Figure 2A,B in Edmunds and Burgess 2016), size-dependent growth only at 25.5 °C and high turbulent flow speed (and not under the other three sets of conditions) is novel. While not as strong in the remaining three relationships of whole-colony calcification against size, a trend for allometry is suggested by the downward depression of the slopes (especially at a low turbulent flow speed; solid lines vs. dashed lines in Fig. 3). Given the less dense sampling in the present study across the same range of colony sizes as used in our earlier analysis (Edmunds and Burgess 2016), it is possible that the underlying size-dependent relationships in the present analysis fundamentally are allometric.

The effect of turbulent flow speed on calcification rate in *Pocillopora verrucosa* depended on the colony size, but in unanticipated ways. Increased turbulent flow speed was expected to alleviate mass transfer limitations within a densely branched colony, removing the effects of size of calcification rate (Falter et al. 2007; Chang et al. 2009; Edmunds and Burgess 2016). Physiological rates are typically reduced in densely versus more loosely branched corals (Lowe et al. 2005; Reidenbach et al. 2006; Falter et al. 2007), but are increased by oscillatory versus unidirectional flow (Reidenbach et al. 2006; Falter et al. 2007). Therefore, it was unexpected that a 204% increase in turbulent flow speed in the current study compared to our previous study (Fig. 2) did not alleviate allometry for calcification (at 25.5 °C, Fig. 3a). However, interpretation of the biological meaning of enhanced mass flux depends on the metabolite, and while the effect is likely to be stimulatory for aerobic respiration at night through the delivery of O_2 (Kühl et al. 1995), photosynthesis through the efflux of O_2 during the day (Mass et al. 2010), and calcification through the delivery of Ca^{2+} and CO_3^{2-} during the day (Dennison and Barnes 1988; Schutter et al. 2010), the effects could be inhibitory if they promoted the loss of a rate-limiting solutes or particles. Such effects have not been investigated in scleractinians, but potentially they could arise through enhanced loss of dissolved organic carbon (Crossland 1987; Brown and Bythell 2005), removal of beneficial bacteria associated with the surface mucous layer of coral tissue (Rosenberg et al. 2007), or perhaps dissipation of ammonium that could be released by the coral tissue in the darkest interstices within the largest coral colonies (Rahav et al. 1989).

Evaluating the above-mentioned possibilities as a means to better understand how calcification in *P. verrucosa* responds to temperature, flow, and colony size, promises to be complex, in part because the effects of these factors on calcification are likely to have indirect components. For instance, our previous work with the same species revealed depressed allometry for gross photosynthesis at low temperature, and enhanced, temperature-dependent allometry for aerobic respiration (Edmunds and Burgess 2016), both of which could indirectly affect calcification by modulating the supply of the organic matrix necessary for mineralization, or the energy resources required for $CaCO_3$ deposition (respectively) (Allemand et al. 2011). Explaining the allometry for calcification for some treatments but not others (Fig. 3a) is likely, therefore, to require a broader focus on the ways in which turbulent flow speed affects respiration, gross photosynthesis, and calcification in large colonies. Such work will benefit from closer attention to the chemical and physical conditions among coral branches under different ambient flow regimes, for example, by measuring flow speeds or its effects (i.e., dissolution) among coral branches (Reidenbach et al. 2006; Chang et al. 2009), or by sampling for pH and oxygen concentration in the diffusion boundary layers surrounding

branching corals (Chan et al. 2016). Measuring flow speeds within branching colonies under different ambient flow conditions is particularly important, as a limitation of the present study is reliance on the inference that a contrast of two turbulent flow speeds affected seawater motion, boundary layers, and the flux of metabolites within the interstices of large, branching coral colonies.

In summary, the present results emphasize the importance of colony size in modulating the calcification rate of a branching coral in response to variation in ecologically relevant turbulent flow speeds and seawater temperature. Regardless of the processes underlying our results, estimates of calcification in small coral colonies (e.g., single branches and fragments) will provide misleading estimates of calcification in larger corals, and of the effects of temperature and turbulent flow speed on most corals in a population. The selective value of colony size is likely to vary among reef locations differing in turbulent flow speed and temperature. Both these environmental factors have the potential to contribute to a better understanding of the distribution of spatial refuges across a coral reef where high rates of coral calcification can be maintained (Kuffner et al. 2013; Castillo et al. 2014; Pratchett et al. 2015), and how future environmental conditions will affect the population structure of reef corals. To realize these possibilities, closer attention will need to be placed on the size distribution of coral colonies on individual reefs, as well as measuring variation in seawater turbulence on spatial scales commensurate with those determining the distribution and growth of coral colonies.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no competing or financial interests.

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Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

Data accessibility Data in support of this manuscript deposited at the MCR-LTER site: http://mcr.lternet.edu/data/ (https://doi.org/10.6073/pasta/d583e9ea5529c11cbc2c2884fd77ae14).

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